The ProbLemma's Channel Season 3 Guide

Season 3 Episode 1: The Index Of Tools

- All the problem-solving approaches studied so far, across seasons 1 and 2, are indexed by the episodes in which they were defined and practiced

Season 3 Episode 2: An Integral Transformation

- Problem S3M1 solved, show that for all θ, x real, with $x > 1$, and $n = 1, 2, 3, 4, 5 \ldots$ it is the case that:

$$\int_0^\pi \left(x + \sqrt{x^2 - 1}\cos(\theta)\right)^n \, \mathrm{d}\theta = \int_0^\pi \frac{\mathrm{d}\theta}{\left(x - \sqrt{x^2 - 1}\cos(\theta)\right)^{n+1}}$$

- Problems S3M2/3/4 formulated:
  - Problems S3M2/3/4: deduce in at least 3 distinct ways an alternative expression (a product) for the following finite sum:

$$\sum_{k=0}^{n} \cos(a + k\theta)$$

Season 3 Episode 3: One Telescope, Two Telescopes

- Problem S3M2 solved

Season 3 Episode 4: A Complex Approach

- Problem S3M3 solved

Season 3 Episode 5: Euclid Says

- Problems S3M4 solved
- Problem S3P1 formulated:
  - S3P1: find the period of oscillation of a triangular spring oscillator

Season 3 Episode 6: Triangular Spring Oscillator

- Problem S3P1 solved
- Problem S3CS1 formulated:
  - S3CS1: sort 3 integers creatively, without the popular containers and/or industrial strength algorithms

Season 3 Episode 7: 3-Integer Sort, A Discovery

- Problem S3CS1 solved

- Problem S3M5 formulated:
  - S3M5: construct a point, N, inside of a right triangle, ABC, such that the magnitudes of all the angles NAB, NBC and NCA are all equal one another

Season 3 Episode 8: Brocard Points Introduction
- Problem S3M5 solved
- Problem S3M6 formulated:
  - S3M6: invent a way to evaluate finite sums based on a popular approach that is used in combinatorial analysis to prove binomial identities

Season 3 Episode 9: Summation By Double-Counting
- Problem S3M6 solved
- Problem S3P2 formulated:
  - S3P2: find the distance covered by a bouncing material point

Season 3 Episode 10: Infinity In Physics
- Problem S3P2 solved
- Problem S3CS2 formulated:
  - S3CS2: suggest a Turing Machine-friendly algorithm for testing if a given point is located outside or inside of a given triangle

Season 3 Episode 11: A Point-in-triangle Detection Algorithm
- Problem S3CS2 solved
- Problem S3M7 formulated:
  - S3M7: find a way to integrate a binomial differential $x^m(a+bx^n)^p\,dx$

Season 3 Episode 12: Integration Of Binomial Differentials
- Problem S3M7 solved
- Problem S3M8 formulated:
  - S3M8: use the integration of binomial differentials machinery developed in the previous problem in order to evaluate the following infinite sum:

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+1)(n+2)\dots(n+p)}, \quad p \geqslant 2$$

Season 3 Episode 13: Binomial Differentials Slay Infinite Sums Before Breakfast
- Problem S3M8 solved
- Problem S3P3 formulated:
  - S3P3: distinguish a sphere from an identical ball of same radius and mass

Season 3 Episode 14: Sisyphus Versus Spherical Chickens Of Uniform Density
- Problem S3P3 solved
- Problem S3CS3 formulated:
  - S3CS3: construct a single-pass algorithm for sorting **0**s and **1**s

Season 3 Episode 15: Single-pass Janus Sort
- Problem S3CS3 solved
- Problem S3M9 formulated:
  - S3M9: construct an equilateral triangle on 3 parallel straight lines

Season 3 Episode 16: An Equilateral Triangle On 3 Parallel Straight Lines Construction
- Problem S3M9 solved
- Problem S3M10 formulated:
  - S3M10: reconstruct a square given 4 points that belong to one side of that square each

Season 3 Episode 17: 4-Point Square Reconstruction
- Problem S3M10 solved
- Problem S3P4 formulated:
  - S3P4: find the normal and the tangential accelerations as functions of time of a point that moves along a parabola with its acceleration vector staying parallel to the y-axis at all times

Season 3 Episode 18: A Parabolic Motion With Acceleration Staying Parallel To The y-axis
- Problem S3P4 solved
- Problem S3CS4 formulated:
  - S3CS4: creatively swap 2 sub-arrays

Season 3 Episode 19: A Creative Swap Of Two Sub-Arrays
- Problem S3CS4 solved
- Problem S3M11 formulated:
  - S3M11: evaluate a finite trigonometric sum

$$\sum_{k=1}^{n} \frac{1}{\sin\left(2^k x\right)}$$

Season 3 Episode 20: A Finite Trigonometric Sum Evaluation
- Problem S3M11 solved
- Problem S3M12 formulated:
  - S3M12: evaluate a finite binomial sum using Physics:

$$\sum_{k=0}^{n} k \cdot \binom{n}{k}$$

Season 3 Episode 21: Physics Evaluates A Finite Binomial Sum
- Problem S3M12 solved
- Problem S3P5 formulated:
  - S3P5: recover the shape of the trajectory of a point that moves in a plane in such a way that its both tangential and normal accelerations remain constant at all times

Season 3 Episode 22: Planar Motion With Constant Tangential And Normal Accelerations
- Problem S3P5 solved
- Problem S3CS5 formulated:
  - S3CS5: explain how the Shunting Yard Algorithm works and create a Slow-Motion Player that lets a user experiment with and witness the mechanics of that algorithm

Season 3 Episode 23: The Mechanics Of The Shunting Yard Algorithm
- Problem S3CS5 solved
- Problem S3M13 formulated:
  - S3M13: evaluate the following integrals in at least 3 different creative ways:

$$C_{a,b}(x) = \int e^{ax} \cos(bx)\, dx; \qquad S_{a,b}(x) = \int e^{ax} \sin(bx)\, dx$$

Season 3 Episode 24: 3 Creative Evaluations Of Ubiquitous Integrals
- Problem S3M13 solved
- Problem S3M14 formulated: a divisibility by 7 test
  - S3M14: design a divisibility-by-7 algorithm using the Reduce-And-Conquer approach

Season 3 Episode 25: A Divisibility-By-7 Test Design Via Reduce-And-Conquer
- Problem S3M14 solved
- Problem S3P6 formulated: a kinematic descend of a rotating line segment
  - S3M14: recover the law of motion, the trajectory, the velocity, the acceleration and the curvature of the trajectory traced by one edge of a uniformly descending line segment that also rotates in the horizontal plane with a constant angular velocity

Season 3 Episode 26: A Kinematic Descend Of A Rotating Line Segment
- Problem S3P6 solved
- Problem S3CS6 formulated: describe the mechanics and then design and implement the Narayana's Next-Linear-Permutation algorithm

Season 3 Episode 27: Narayana's Next Linear Permutation Algorithm
- Problem S3CS6 solved
- Problem S3M15 formulated. Evaluate the following finite trigonometric sum:

$$\sum_{m=1}^{n} \arctan\left(\frac{2m}{m^4 + m^2 + 2}\right)$$

Season 3 Episode 28: In Like A Lion Out Like A Lamb
- Problem S3M15 solved
- Problem S3M16 formulated. Evaluate the following limit:

$$\lim_{n\to+\infty} n^{\frac{1}{n}}$$

Season 3 Episode 29: The Art Of Manipulation Of Inequalities
- Problem S3M16 solved
- Problem S3P7 formulated: deduce an equation that describes a spherical motion of a point under a constant angle with the meridians

Season 3 Episode 30: A Deduction Of An Equation Of A Loxodrome
- Problem S3P7 solved
- Problem S3CS7 formulated: construct an $O(n)$ time and $O(1)$ space complexity algorithm for detecting if a given integer can be represented as a sum of two integers from a given array sorted in some order, ascending or descending

Season 3 Episode 31: An Integer As A Sum Of Two Elements Of A Sorted Array
- Problem S3CS7 solved
- Problem S3M17 formulated: evaluate the following indefinite integral integral

$$\int \sqrt{x^2 - a^2}\, dx$$

in at least two creative ways.

Season 3 Episode 32: Hyperbolically Speaking
- Problem S3M17 solved
- Problem S3M18 formulated: evaluate the following definite integrals

$$\int_0^{\frac{\pi}{2}} \sin^m(x)\, dx \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^m(x)\, dx$$

for $m = 1, 2, 3, 4, 5$ and so on.

Season 3 Episode 33: Recurrence Relations Evaluate Integrals
- Problem S3M18 solved
- Problem S3M19 formulated: show that hyperbolic functions can be defined by a direct analogy with circular functions defined via origin-centered unit circle

Season 3 Episode 34: Hyperbolic Functions Defined By A Direct Analogy With Circular Functions
- Problem S3M19 solved
- Problem S3P8 formulated: recover the law of motion of a point that is moving along a straight line that sweeps the surface of a right circular cone

Season 3 Episode 35: Twice Telescopic Rotating Antenna, The Kinematics Of
- Problem S3P8 solved
- Problem S3CS8 formulated: construct an algorithm for detecting a cycle in a linked list

Season 3 Episode 36: A Vicious Circle Or Lords Of The Ring
- Problem S3CS8 solved
- Problem S3M20 formulated: show that for all real $x \neq \pm k\pi$, where $k = 0, 1, 2, 3$ and so on, it is the case that:

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Season 3 Episode 37: Sine As A Product, With Applications
- Problem S3M20 solved
- Problem S3M21 formulated: evaluate the Euler-Poisson integral

$$\int_0^{+\infty} e^{-x^2}\, dx$$

using the Squeeze Theorem.

Season 3 Episode 38: What Mathematicians Do All Day Or Euler-Poisson Integral Evaluation via the Squeeze Theorem
- Problem S3M21 solved
- Problem S3P9 formulated: solve the S2E54/S2P2 problem under the assumption that there is no external input into the rotating rod-bead system, which means that the system's angular velocity is no longer constant.

Season 3 Episode 39: A Bead Sliding Off Of A Massless Rod That Rotates With A Decreasing Angular Velocity
- Problem S3P9 solved
- Problem S3CS9 formulated: find the diamond across 8 boxes with 2 cooperating people if boxes have a random orientation, the two people must agree on the algorithm ahead of time, one person is lead into the room with the boxes to do something and then escorted out of the room, while the second person, upon entering that room, has to recover the box with the diamond

Season 3 Episode 40: How Computer Science And Group Theory Found A Diamond

- Problem S3CS9 solved
- Problem S3M22 formulated: solve the following linear non-homogeneous recurrence relation:

$$t_n = 6t_{n-2} - t_{n-1} - 4, \quad t_0 = 2, t_1 = 4$$

Season 3 Episode 41: How Generating Functions Can Solve Linear Non-Homogeneous Recurrence Relations

- Problem S3M22 solved
- Problem S3M23 formulated: how many chords are formed by n distinct points located on the circumference of a circle and why?

Season 3 Episode 42: Chords Of The Ring

- Problem S3M23 solved
- Problem S3P10 formulated: two points execute a coplanar motion as follows. A point R moves along a straight line with constant velocity $\overrightarrow{v_r}$, while a point C moves (in the same plane) with a velocity, $\overrightarrow{v_c}$, that is constant by magnitude but always points at the point R. Find the trajectory of the point C, the condition and the time of the intercept of R by C, the intercept time from the moment when the vectors $\overrightarrow{v_r}$ and $\overrightarrow{v_r}$ were perpendicular and the distance $|CR|$ was d_0.

Season 3 Episode 43: The Kinematics Of Wile E. Coyote and Road Runner

- Problem S3P10 solved
- Problem S3CS10 formulated: invent/discover/construct/synthesize/put together/come up with a practically useful $O(1)$ time complexity algorithm.

Season 3 Episode 44: A Story Of An $O(1)$ Bucket List

- Problem S3CS10 solved
- Problem S3M24 formulated: solve the equation $z \cdot z = \overline{z}$ over $\mathbb{C}$

Season 3 Episode 45: There Is More To Mathematics Than Meets They Eye

- Problem S3M24 solved
- Problem S3M25 formulated: find the shape of an orbit from each point of which both the Earth and the Moon are seen under one and the same angle

Season 3 Episode 46: Where Problems Live Or Earth-Moon Equiangular Orbit

- Problem S3M25 solved
- Problem S3P11 formulated: find the velocity of a point that orbits a circle in such a way that its areal velocity with respect to an off-center point located strictly inside of that circle is constant as a function of the distance from that off-center point to the moving point itself

Season 3 Episode 47: A Hidden Kepler Or Isomorphism In Celestial Mechanics

- Problem S3P11 solved

- Problem S3CS11 formulated: implement a thread-safe blocking $O(1)$ queue following the sprit of the solution of the problem S3CS10, ensuring that this queue exhibits the following behavior:
  - it blocks if it is full on a call to `qAdd()` and
  - it blocks if it is empty on a call to `qRm()`

Season 3 Episode 48: A Thread-Safe Blocking $O(1)$ Add/Remove Queue
- Problem S3CS11 solved
- Problem S3M26 formulated: show that no non-trivial harmonic sum $H_{n>1}$:

$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

is ever an integer.

Season 3 Episode 49: A Harmonic Argument
- Problem S3M26 solved
- Problem S3M27 formulated: explain why $\zeta(1)$ diverges but $\zeta(2)$ converges.

Season 3 Episode 50: Why $\zeta(1)$ Diverges But $\zeta(2)$ Converges
- Problem S3M27 solved
- Problem S3P12 formulated: determine the velocity and the acceleration of a point moving over an origin-centered ellipse, with semi-axes a and b, with a constant velocity, v_0, as functions of the point's coordinates, x and y.

Season 3 Episode 51: A Uniform Motion Over An Ellipse, A Verbal Exam Scenario
- Problem S3P12 solved
- Problem S3CS12 formulated: construct an $O(n)$ algorithm for reversing the direction of a singly-linked list to its opposite in a single pass.

Season 3 Episode 52: Hansel And Gretel Do Comp Sci
- Problem S3CS12 solved
- Problem S3M28 formulated: decompose the following real-to-real single-variable functions:

$$\cot(x), \quad \tan(x), \quad \frac{1}{\sin(x)}, \quad \frac{1}{\sin^2(x)}$$

into the corresponding partial fraction series using the machinery of Real Analysis only.

Season 3 Episode 53: A Real Cornucopia Of Partial Fraction Decompositions
- Problem S3M28 solved
- Problem S3M29 formulated: evaluate the following improper integral:

$$\int_0^{+\infty} \frac{\sin(x)}{x}\, dx$$

using the decompositions obtained in S3M28/S3E53.

Season 3 Episode 54: Partial Fraction Decompositions Evaluate Integrals
- Problem S3M29 solved
- Problem S3P13 formulated: determine the trajectory of a ladybug sitting on a rake that slides down a barn floor in such a way that the top of the rake's handle remains in contact with the barn's wall at all times.

Season 3 Episode 55: Kinematics In A Barn Or A Ladybug On A Rake
- Problem S3P13 solved
- Problem S3CS13 formulated: construct an $O(n)$ algorithm for determining the output position of an arbitrary input square of a 2^n-strip of paper after that strip was folded in half onto itself n times

Season 3 Episode 56: Strip Teez
- Problem S3CS13 solved
- Problem S3M30 formulated: prove the relevant theorems for the Frullani Integrals

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x}\, dx$$

Season 3 Episode 57: Frullani Integrals Forever
- Problem S3M30 solved
- Problem S3M31 formulated: show how improper integrals of rational functions can be evaluated

$$\int_{-\infty}^{+\infty} \frac{P_m(x)}{Q_n(x)}\, dx,\ m+2 \leqslant n$$

Season 3 Episode 58: The Real Granddaddy of the Cauchy's Residue Theorem
- Problem S3M31 solved
- Problem S3PS14 formulated: find the trajectory of a point on the circumference of a circle of radius r rolling without sliding and spinning over 1) the exterior and 2) the interior of a fixed circle of radius R

Season 3 Episode 59: Epi/Hypo/Cycloids, Copernicus, Ladybug, Chicken & A Vacuum Cleaner
- Problem S3P14 solved
- Problem S3CS14 formulated: describe the structure of the fixed points from the Strip Teeze computer science problem S3CS13

Season 3 Episode 60: The Structure Of Fixed Points Of Strip Teeze
- Problem S3CS14 solved
- Problem S3M32 formulated: establish the less-than/equal-to/greater-than relationship between the following two single-variable real-to-real functions, $\cos(\sin(x))$ and $\sin(\cos(x))$

Season 3 Episode 61: Balance? Or Tip? Which Way? And Why?
- Problem S3M32 solved
- Problem S3M33 formulated: prove without using integration:

$$\pi = 2 \cdot \int_{-1}^{1} \sqrt{1 - x^2}\, dx$$

Season 3 Episode 62: Prove Without Evaluating Integrals
- Problem S3M33 solved
- Problem S3P15 formulated: find the trajectory, the velocity and the acceleration of a point executing a planar motion in such a way that the product of distances from that point to two fixed, distinct coplanar points is equal to the square of half-distance between the said fixed points

Season 3 Episode 63: A Kinematics Of Lemniscate Of Bernoulli
- Problem S3P15 solved
- Problem S3CS15 formulated: design and implement an algorithm that will print out all the subsets of a set of n numbers, where n is fixed and given ahead of time

Season 3 Episode 64: Print All Subsets Of A Finite Set Of Size n in $O(n \cdot 2^n)$ Time
- Problem S3P15 solved
- Problem S3M34 formulated: evaluate the following integral creatively:

$$\int_{0}^{\pi/2} \frac{x^2}{(x \sin x + \cos x)^2}\, dx$$

Season 3 Episode 65: Integrate By Differentiation
- Problem S3M34 solved
- Problem S3M35 formulated: construct an ordinary power series generating function for the sequence of Fibonacci numbers $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$

Season 3 Episode 66: OPSGFs Vs Fibonacci Numbers Or Mathematics Is A Process
- Problem S3M35 solved
- Problem S3P16 formulated: two points, a runner and a chaser, execute their motions in the same plane in such a way that the runner moves in a straight line and the vector of the chaser's velocity always points at the runner and the distance between the runner and the chaser remains constant at all times. Find the trajectory of the chaser, its law of motion, its acceleration if that acceleration is always orthogonal to its velocity; find the law of motion of the runner

Season 3 Episode 67: Dominos, Perrault, Lobachevsky And A Rake
- Problem S3P16 solved
- Problem S3CS16 formulated: the lights in an unknown number of rooms positioned in a circle in an underground tunnel form a random ON/OFF pattern. Find an $O(n)$ algorithm for counting the number of rooms in such a configuration with absolute certainty by using only the binary, ON/OFF, light switches installed in each room

Season 3 Episode 68: The $O(n)$ Light At The End Of A Circular Tunnel
- Problem S3CS16 solved
- Problem S3M36 formulated: evaluate the following finite binomial sum algebraically:

$$\sum_{k=0}^{n} \binom{n+k}{n-k}$$

Season 3 Episode 69: OPSGFs Evaulate Finite Binomial Sums
- Problem S3M36 solved
- Problem S3M37 formulated: evaluate the same sum combinatorially

$$\sum_{k=0}^{n} \binom{n+k}{n-k}$$

Season 3 Episode 70: Combinatorix Evaulates Finite Binomial Sums
- Problem S3M37 solved
- Problem S3P17 formulated: determine the trajectory of a point that moves with a constant linear velocity relative to a circle that itself rotates in the parent plane about a point on its circumference

Season 3 Episode 71: Kinematic Rozez
- Problem S3P17 solved
- Problem S3CS17 formulated: print all prime numbers between 2 and 100 without using literal numbers and without using ASCII symbols

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